MEET BHATT

mbbhatt@andrew.cmu.edu — (380) 226-4073 — meetbhatt.com — GitHub — LinkedIn

EDUCATION

Carnegie Mellon University – Pittsburgh, PA — December 2027
Master of Science in Artificial Intelligence Engineering and Technology Innovation Management

Pandit Deendayal Energy University (PDEU) – Gujarat, India — June 2026
Bachelor of Technology in Computer Engineering, Minor in Internet of Things — GPA: 9.55/10

AWARDS & HONORS

Overall Grand Winner – 7th Global AI Hackathon by Hack-Nation (Vitamin Bob team)
Health Track Winner – World Bank Group Small AI for Development Hackathon Challenge (Vitamin Bob team)
Invited to present at the WBG Global AI & Digital Summit in Seoul.
J N Tata Scholarship — *Scholar and scholarship recipient*

SKILLS

Programming & Data: Python, SQL, pandas, Polars, Pandera, scikit-learn
Machine Learning & Deep Learning: PyTorch, TensorFlow/Keras, ARIMA/SARIMA, Random Forest, XGBoost, CNNs, RNNs, Time Series Analysis
Visualization & Explainability: Tableau, MLflow, SHAP, Grad-CAM
Generative AI: Hugging Face Transformers, RLHF, supervised fine-tuning, LLM evaluation, prompt classification
Platforms & Deployment: Google Cloud Platform, Vertex AI, Docker, Kubernetes, Supabase, Firebase, GitHub Actions

PROFESSIONAL EXPERIENCE

Health and Human Performance Lab, Carnegie Mellon University — September 2026 – December 2026
Research Assistant — Advisors: Prof. Kirk Warren Brown and Prof. J. David Creswell
Skills: Intensive longitudinal analysis, treatment-response prediction, interpretable ML, cross-validation
- **Applying machine learning to longitudinal data** to predict individual outcomes of mindfulness-based interventions.
- **Extending interpretable, cross-validated prediction** from pre/post designs to daily-life measurements to identify participant and practice patterns linked to treatment benefit.

Corporate Startup Lab – Bosch Mobility — August 2026 – December 2026
Skills: Market research, competitive analysis, strategic-fit assessment, customer discovery
- **Co-lead the Human Augmentation segment** of a six-person team, evaluating how Bosch's low-voltage actuators could enter the occupational exoskeleton market without major re-engineering.
- **Run right-to-fit analyses** on prospective customers against product and regulatory requirements, and mapped Bosch's **seven actuator product lines** using primary-source data.

Blink Analytics — serin-ai.com — August 2024 – June 2026
RLHF Contributor → RLHF Team Lead → Lead Product Development Intern
Skills: Multimodal scoring, RLHF operations analytics, LLM evaluation, quality assurance
- **Designed Serin's candidate evaluation framework**, combining multimodal signals into three scoring indices used to rank candidates across **500+ interviews**.
- **Evaluated LLM outputs and led 10-person RLHF workflows**, building throughput and label-quality tracking plus review rubrics; insights helped drive a **222% revenue increase in two months**.

RESEARCH EXPERIENCE

Project Phoenix – Explainable Cervical Cancer Classification — July 2025 – May 2026
Website — GitHub — *Manuscript in preparation* — **Skills:** Image processing, CNNs, Grad-CAM
- **Benchmarked four image-enhancement pipelines** on 4,049 SipakMed images across five classes using PSNR, SSIM, edge-preservation, and contrast metrics; **NLM+CLAHE scored highest** (edge preservation: 0.60 vs. 0.37 for NLM).
- **Fine-tuned and compared three CNN architectures** on a 3,239/405/405 split; ConvNeXtV2 achieved **96.3% accuracy, 0.963 macro-F1, and 0.996 ROC-AUC**. Per-class analysis flagged Koilocytotic and Metaplastic as the weakest classes (F1 $\approx$ 0.93); Grad-CAM was used to inspect regions driving predictions.

Malicious Prompt Classifier – Markov Chain Detection System — July 2025 – October 2025
GitHub — Published paper — **Skills:** Markov chains, NLP, interpretable classification
- **Built an interpretable Markov-chain classifier** on **39,234 balanced prompts**, using likelihood-ratio scoring to surface the word sequences behind each flag.
- Achieved **90.79% accuracy, 98.84% precision, 82.54% recall, and 89.96% F1**; retained the simpler baseline after a higher-order ensemble reached only 89.3% accuracy.

PROJECTS

Vitamin Bob – Offline AI Patient Triage — October 2026
GitHub — Demo — **Skills:** Python, FastAPI, Gemma 4 E2B, llama.cpp
Overall Grand Winner, 7th Global AI Hackathon; Health Track Winner, WBG Small AI for Development Challenge
- **Built offline missed-call triage and callbacks** with FastAPI; Gemma 4 E2B (llama.cpp) structures Hindi/Gujarati speech for rule-based routing to emergency services, clinics, or appointments.
- **Ran grammar-constrained inference in 5–7 seconds** on a 4 GB laptop GPU; across 64 tests, missed zero urgent cases and halved uncertainty versus a keyword baseline. Added keypad fallback, SMS, encrypted sync, and a clinician dashboard.

Linea – Desktop Music Lyrics Overlay — April 2026 – Present
Website — GitHub — **Skills:** Electron, real-time integration, privacy-first design
- **Built a transparent, always-on-top Electron lyrics overlay** with time synchronization, click-through controls, playback integration, and **local-only storage**.

Project Talaria – Wearable Cardiovascular & Gait Monitor — July 2025 – November 2025
Website — GitHub — **Skills:** pandas, LSTM, TensorFlow Lite, Vertex AI
- **Cleaned and merged 143K sensor records from 32 participants**, resolving type errors, imputing missing values, and standardizing 15 physiological and motion features.
- **Built 50-timestep sequences and an LSTM next-step predictor**, converted it to **TensorFlow Lite** (0.19 MB, <1% deviation from the full Keras model), and served predictions through **Vertex AI** to a live dashboard.

LEADERSHIP

Mind Ripple — The Quizzing Club of PDEU — *President → Advisor* — November 2022 – April 2026
- **Led 30 members**; grew Matrix Breakout to **300+ participants** and increased earnings **10% year over year**.

Interact Club of Baroda Sayajinagari — *Charter President* — July 2021 – June 2022
- **Founded and led a 32-member youth service club** partnering with local nonprofits.